PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS

39th ANNUAL CONVENTION

Fueling Progress: Shaping Animal Nutrition for a Competitive and Sustainable Future

6 OCTOBER 2026

Grand Ballroom, Okada Manila











PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS
39th Annual Convention | October 6, 2026

Optimum Nutrient Levels and Stronger Margins in Modern Corn-Soy Feed Formulation

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Challenges driving the need for nutritional optimization



Growth models and optimizers can guide nutritional decisions



Theories behind the model…

Broiler Growth Model

- **Potential growth** can be predicted when **genotype** is adequately described
- **Desired feed intake** can be predicted from this potential growth and the feed offered.
- **Actual feed consumed** can be predicted, taking the **constraints** on intake into account (e.g. bulk, heat)
- **Actual growth** can be predicted from this actual feed intake.

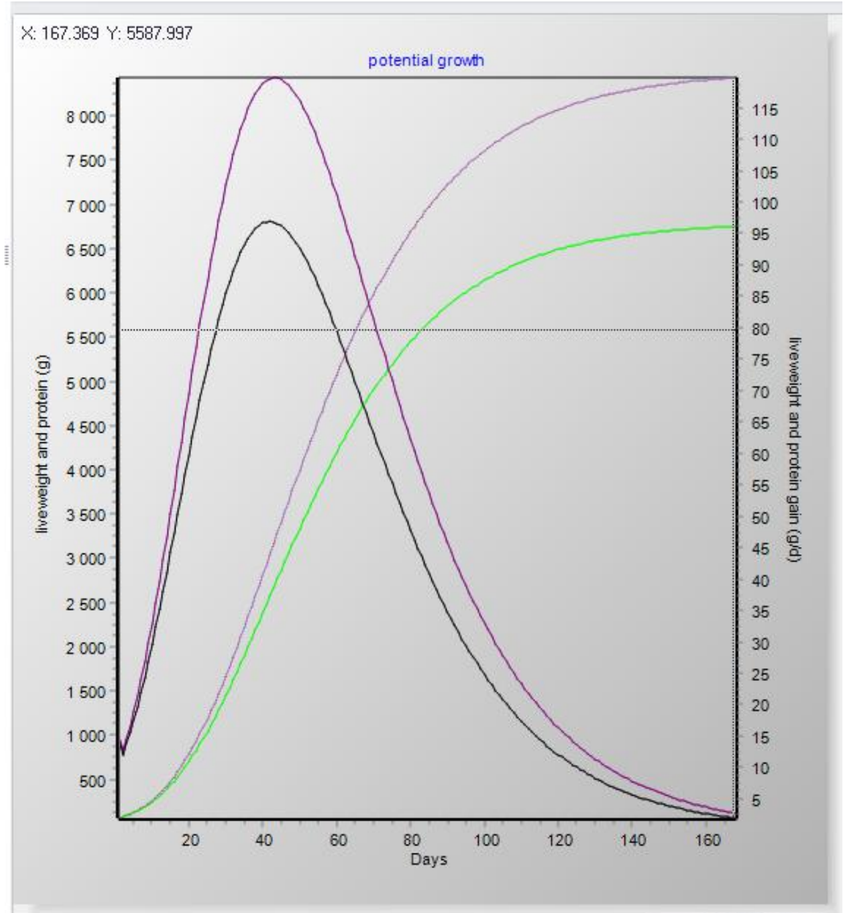


Salisbury (2025)

Potential growth of body protein is described by the Gompertz growth curve

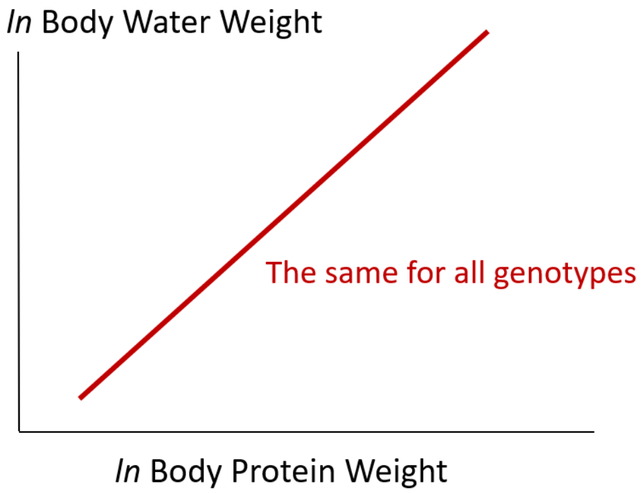
Components of Gompertz Growth Curve:

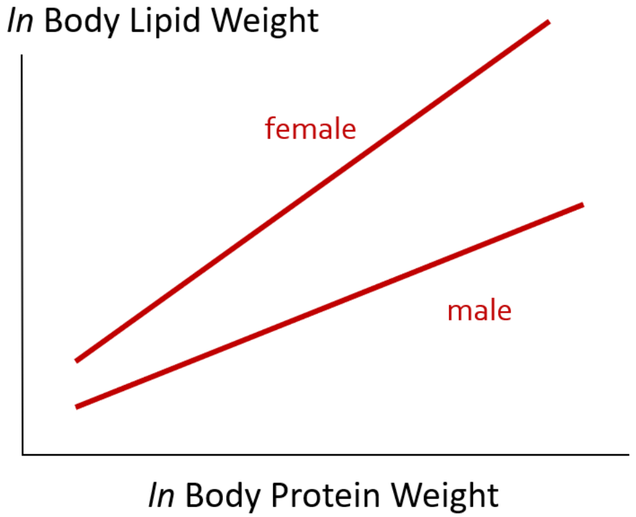
Initial weight	A_0 , kg
Mature weight	A_m, kg
Rate of maturing	B, /d

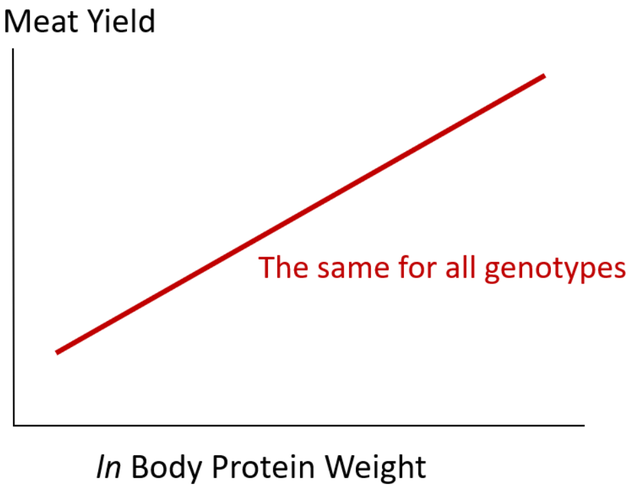


Emmans (1989)

Potential growth of body protein other chemical components and body parts are predicted through allometry to body protein







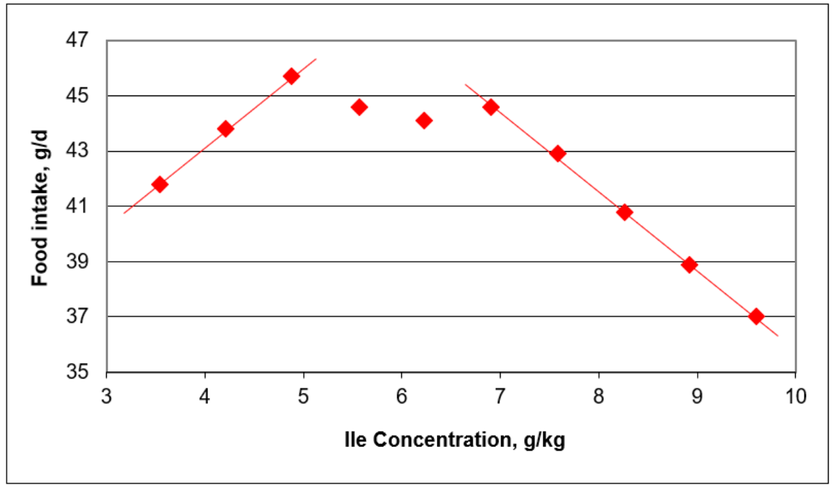
Emmans (1989), Salisbury (2025)

Feed intake is predicted based on the limiting nutrient

Assumption by Emmans (1981)

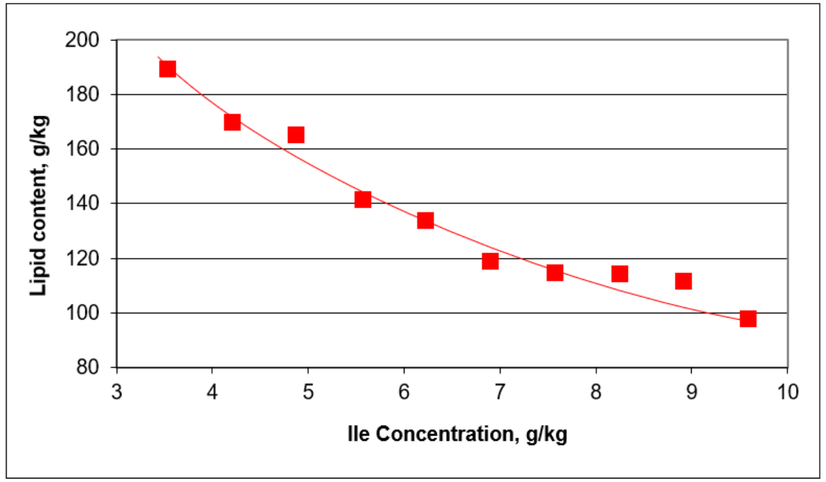
As the concentration of the limiting nutrient in the feed is reduced, food intake will increase in an attempt by the bird to compensate for the deficiency

Evidence



Food intake of chicks fed a range of Ile-limiting feeds

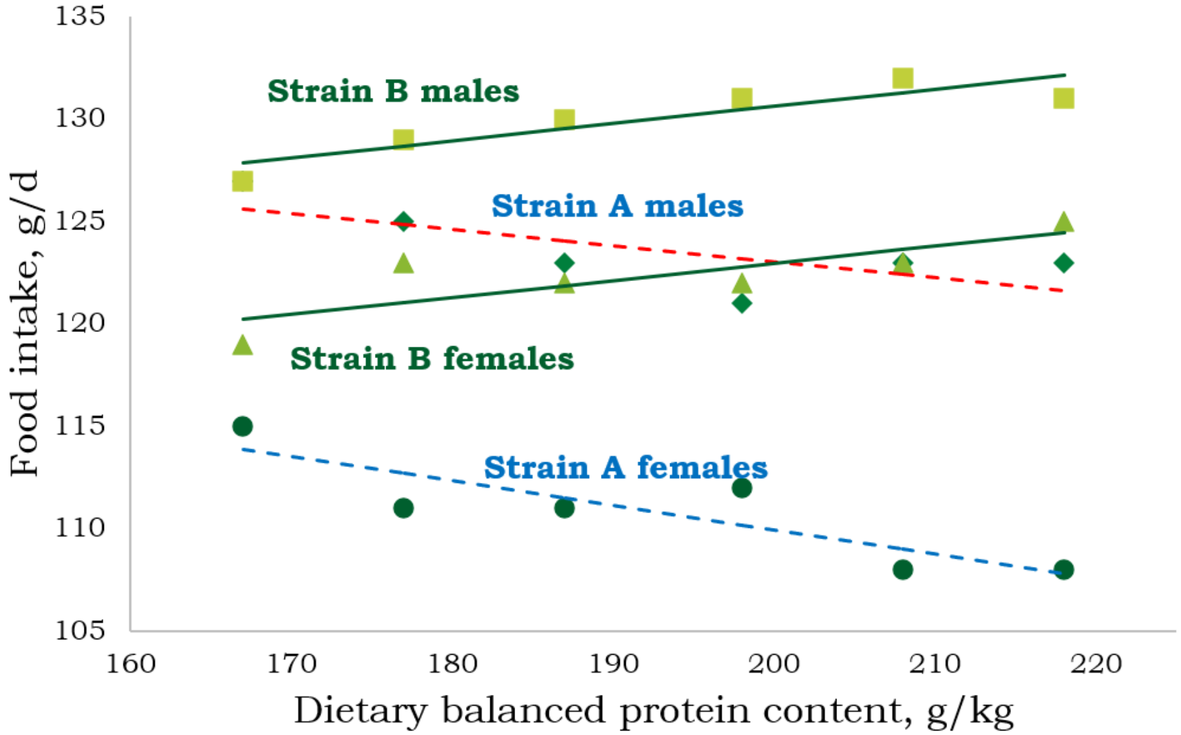
Consequence



Lipid content of chicks fed a range of Ile-limiting feeds

Kemp et al. (2005), Gous (2025)

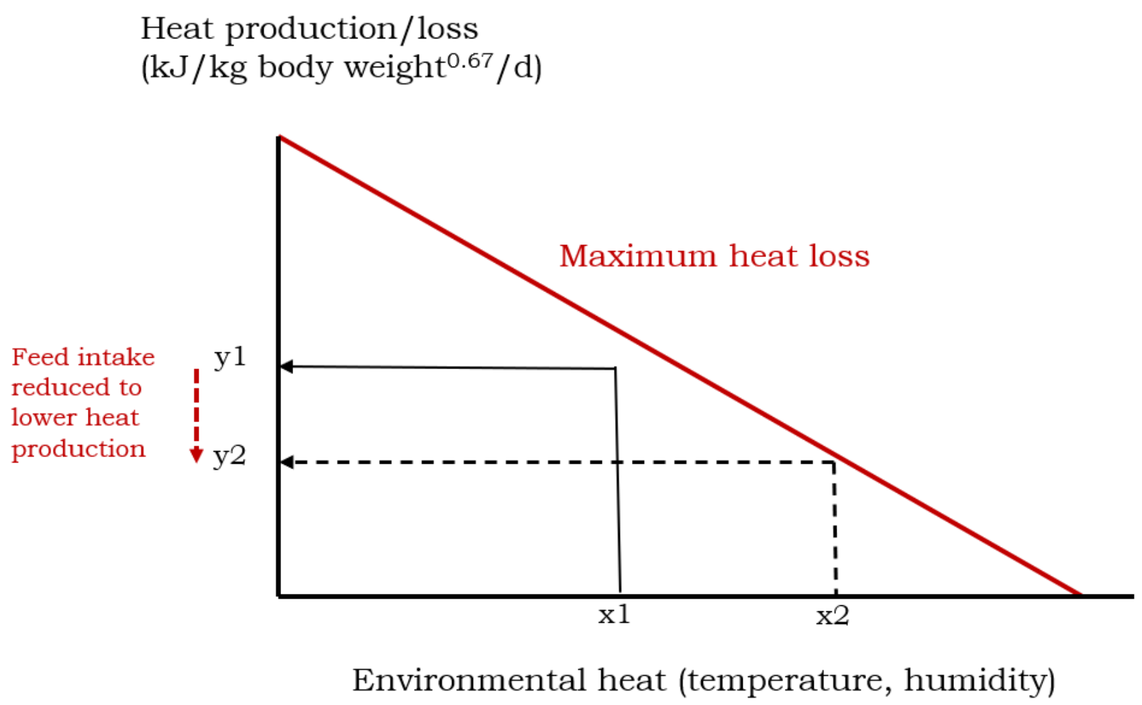
Feed intake differs by sex and genotype



Berhe and Gous (2008); Azevedo et al. (2021)

Constraints on feed intake

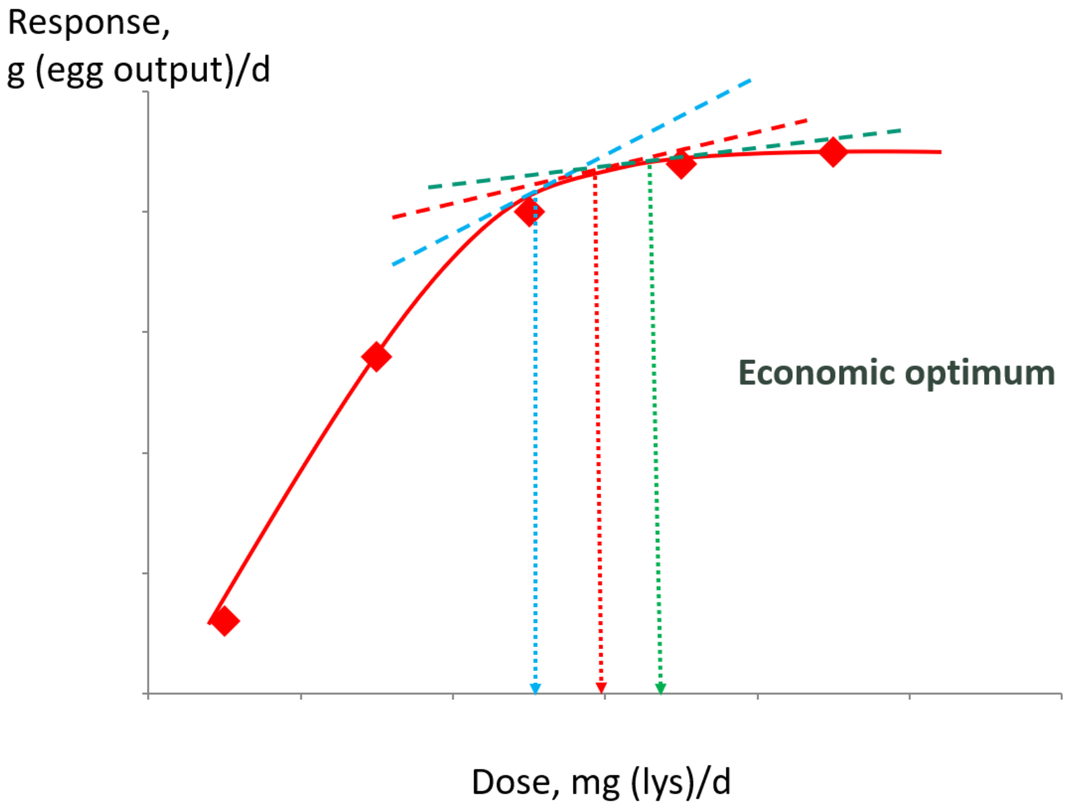
Heat



Bulk - If the feed exceeds the bulk capacity of the bird, it will reduce feed intake.

Salisbury (2025)

Economic Optimizer:
Economic optimum depends on marginal cost of input and marginal return of output



Reading Model, Fisher et al. (1973)

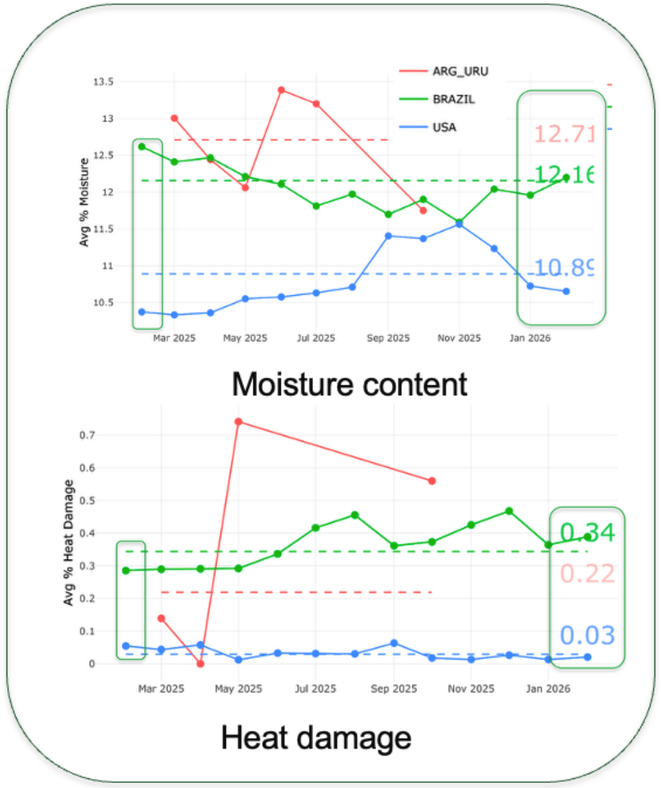
Application 1: Diet Optimization

	Base	Minimize FCR	Maximize margin over feed cost
Feed	Lysine content, g/kg		
Starter	12.8	14.0	13.2
Grower	11.8	12.8	11.5
Finisher	10.0	12.0	8.0
Performance variable			
Live weight at 35 d (g)	2020	2007	2102
Feed intake to 35 d (g)	3507	3417	3725
FCR (g feed: g gain)	**1.58**	**1.55**	**1.61**
N excretion (g/bird)	9.99	10.5	9.60
Cost of feeding (c/bird)	1622	1620	1620
Revenue (c/bird)	2224	2208	2331
Margin over feed cost (c/bird)	**602**	**588**	**711**
Cost of production (c/kg)	1094	1101	1043

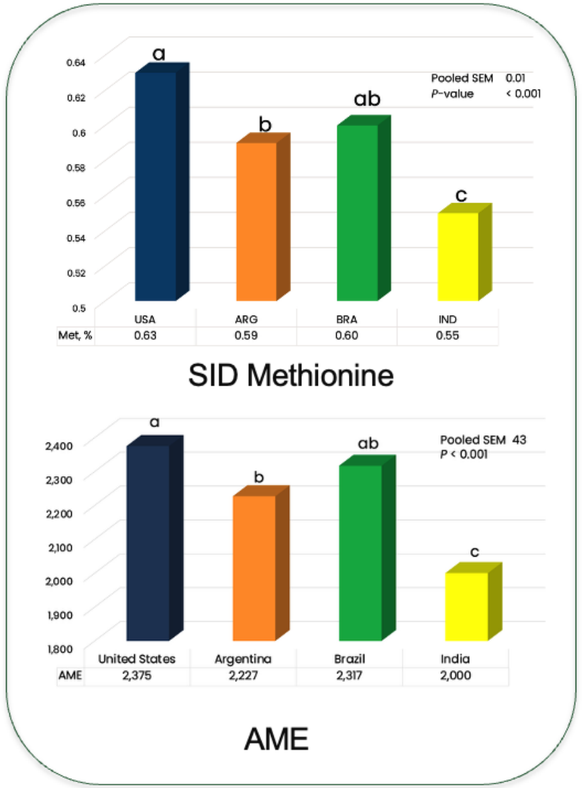
Gous (2025)

Application 2:
Strategic Ingredient Selection

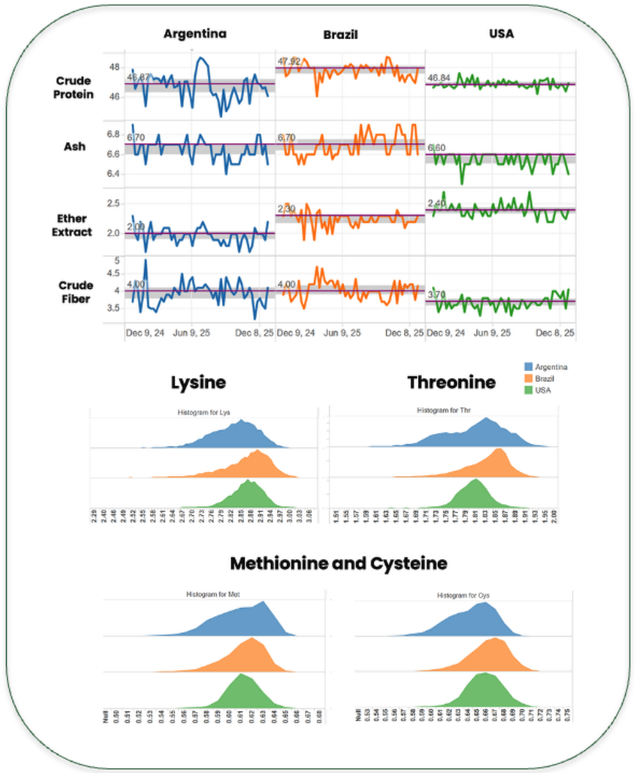
Case: Soybean Meal from Different Origins



Soybean Quality



SBM Nutrient Bundle



SBM Consistency

Ravindran et al. (2014); AgCommodities (2026); Evonik (2026)

Application 2:
Strategic Ingredient Selection

	US SBM		ARG SBM		BRA SBM	
	Base	Opt.	Base	Opt.	Base	Opt.
Feed	AME, MJ/kg					
Phase 1	12.4		12.4		12.4	
Phase 2	12.8		12.8		12.8	
Phase 3	13.0		13.0		13.0	
Performance variable						
Live weight , g	2035		2031		2029	
Harvest age, days	33		33		33	
Feed intake, g	3467		3420		3419	
FCR (g feed: g gain)	1.549		1.531		1.532	
Cost of feeding (c/bird)	177		181		179	
Revenue (c/bird)	400		400		399	
Margin over feed cost (c/bird)	**223**		**219**		**221**	
Cost of production ($/kg)	1.69		1.72		1.70	

Even after optimizing to maximize profit, using U.S. soybean meal gives an additional margin of up to USD 50,000 per 1 million broilers

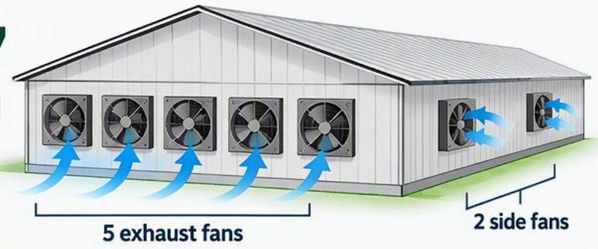
Performance simulated with the same genetics, environmental condition, available ingredients and prices, diet specifications, and product prices

Validation:

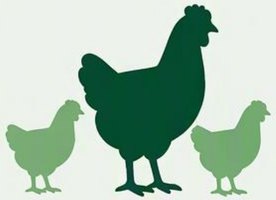
Research trial in a research facility of a big broiler integrator in Indonesia

House profile

H7
Closed house



Animals



4,851
Straight-Run Fast-Feathered Broilers

Treatments

Treatment 1



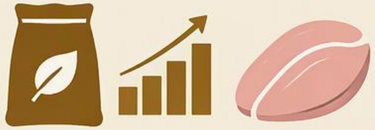
Breed Recommendation

Treatment 2



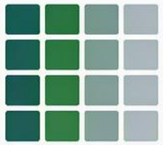
Optimized Diets to Maximize MOFC

Treatment 3



Optimized Diets to Maximize Breast Meat Yield

Experimental design



Randomized Complete Block Design



21
Replicate Pens



77
Birds per Pen

Validation:

Research trial in a research facility of a big broiler integrator in Indonesia

Experimental Diets

PRE-STARTER (0 – 10 DAYS)

	AMEn (kcal/kg)	Dig. Lys (%)	Feed Cost ($/MT)
T1	2976	1.26	492.77
T2	2959 ↓ 17	1.25 ↓ 0.01	489.91 ↓ 2.86
T3	3064 ↑ 88	1.30 ↑ 0.04	507.38 ↑ 14.61

STARTER (11 – 21 DAYS)

	AMEn (kcal/kg)	Dig. Lys (%)	Feed Cost ($/MT)
T1	3050	1.14	490.18
T2	2999 ↓ 51	1.12 ↓ 0.02	482.28 ↓ 7.90
T3	2925 ↓ 125	1.09 ↓ 0.05	471.15 ↓ 19.03

FINISHER (22 – 35 DAYS)

	AMEn (kcal/kg)	Dig. Lys (%)	Feed Cost ($/MT)
T1	3100	1.06	485.58
T2	3023 ↓ 77	1.03 ↓ 0.03	473.29 ↓ 12.29
T3	3119 ↑ 19	1.07 ↑ 0.01	488.52 ↑ 2.94

T1 = Breed Recommendation
T2 = Maximize MOFC
T3 = Maximize Breast Meat Yield

Validation:

Research trial in a research facility of a big broiler integrator in Indonesia

Production Performance (0-35 days)

TREATMENT	DOC BW (g)	BW 35 DOA (g)	ADG (g)	Feed Intake (g)	FCRmc	FCR	IP	IPmc	MOR (%)	CULL (%)	Dep (%)
T1: Breed Recommendation	45.41	2462	69.03	3608	1.466	1.473	471	473	0.93	0.50	1.43
T2: MOFC	45.41	2467	69.18	3594	1.458	1.466	473	475	1.18	0.56	1.73
T3: Breast Meat Yield	45.41	2463	69.08	3586	1.456	1.464	473	475	1.30	0.37	1.67
***P* value**	1.000	0.971	0.972	0.709	0.124	0.233	0.915	0.887	0.634	0.804	0.729
SEM	0.037	8.917	0.255	10.944	0.002	0.003	2.485	2.324	0.160	0.115	0.199

No difference in performance

Validation:

Research trial in a research facility of a big broiler integrator in Indonesia

Breast Yield, %

TREATMENT		Live weight (g)	Carcass (%)	Breast (%)
T1: Breed Recommendation	Male	2,463^{ab}	74.54	30.99^{ab}
	Female	2,444^{ab}	75.17	32.43^{ab}
T2: MOFC	Male	2,478^{ab}	74.13	30.25^{b}
	Female	2,425^{b}	75.07	32.57^{a}
T3: Breast meat yield	Male	2,485^{a}	74.08	30.72^{ab}
	Female	2,432^{ab}	74.50	31.83^{ab}
***P* value**			0.010	0.650	0.017
SEM			6.035	0.196	0.244

^{a,b,c,} means with the same superscripts are not significantly different at $P<0.05$
Carcass (%) : Eviscerated carcass (without neck, feet, and internal organs) as a percentage of live weight.
Breast (%) : Breast meat (with skin and bone) as a percentage of live weight.



No difference between T1 and T3

Validation:

Research trial in a research facility of a big broiler integrator in Indonesia

Feed Cost/kg

TREATMENT	Feed Cost/kg weight gain				
	0-10 d	11-21 d	22-35 d	0-35 d	
T1: Breed Recommendation	IDR 9,418[a] **(USD 0.538)**	IDR 11,036 **(USD 0.631)**	IDR 14,422[a] **(USD 0.824)**	IDR 12,743[a] **(USD 0.728)**	
T2: MOFC	IDR 9,230[b] **(USD 0.527)**	IDR 10,998 **(USD 0.628)**	IDR 13,850[c] **(USD 0.791)**	IDR 12,400[c] **(USD 0.709)**	**Savings of USD 0.019/kg**
T3: Breast Meat Yield	IDR 9,564[a] **(USD 0.546)**	IDR 10,922 **(USD 0.624)**	IDR 14,164[b] **(USD 0.809)**	IDR 12,591[b] **(USD 0.719)**	**Savings of USD 0.009/kg**
***P* value**	0.0003	0.129	<0.0001	<0.0001	
SEM	35.740	23.495	47.281	25.515	

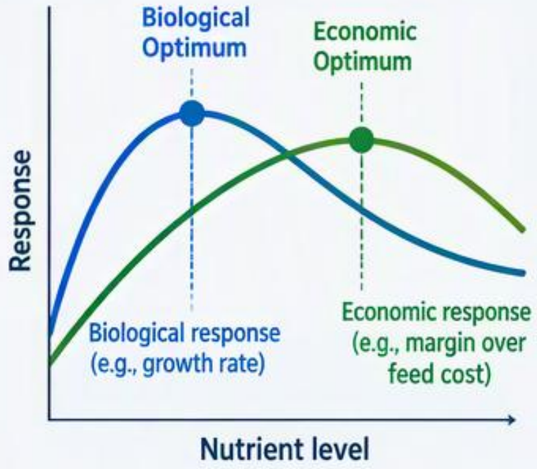
[a,b,c,] means with the same superscripts are not significantly different at $P<0.05$
IDR - Indonesian Rupiah
Pre-starter (0-10d) feed price T1 = Rp 8,624, T2 = Rp 8,573, and T3 = 8,879
Starter (11-21d) feed price T1 = Rp 8,575, T2 = Rp 8,434, and T3 = Rp 8,245
Finisher (22-35d) feed price T1 = Rp 8,498, T2 = 8,283, and T3 = Rp 8,549
1 USD = Rp 17,500

- ✓ **Lower feed cost/kg in optimized diets**
- ✓ **When translated to margins, additional margin of up to USD 19,000 per 1,000 MT chicken meat produced**

Key Takeaways

Biology vs Economics

Different optima for different objectives



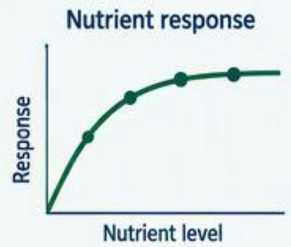
Economic optimum nutrient levels differ from biological optimum nutrient levels.

Predictive Nutrition

From biology to actionable insights

- Genotype
- Feed intake
- Environmental constraints
- Nutrient response





Mechanistic growth models enable prediction of biological responses across nutritional scenarios.

From Formulation to Performance

Optimize nutrients for the best return

Nutrient	Optimized level (e.g.)
AMEn (kcal/kg)	↓ (optimized)
Digestible Lysine (%)	(optimized)
Other amino acids	(balanced)
Feed cost ($/MT)	↓
Performance (BW, FCR, Yield)	=

Maintain performance

Lower feed cost

Precision in nutrient allocation can reduce feed cost without compromising production performance.

Feeding Value Matters

Ingredient quality influences the attainable economic optimum

High-quality soybean meal

- Higher digestibility
- Greater nutrient availability
- Improved animal response
- Higher economic value

Ingredient quality affects the attainable economic optimum.

Thank you!

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